SARIMA Framework

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The Challenge of Seasonal Data

- Time series data often exhibits predictable, repeating patterns.
- These seasonal patterns, along with long-term trends, violate the stationarity assumption of standard ARIMA models.
- How do we model and forecast a series with both non-seasonal and seasonal components?
- Solution: Seasonal ARIMA, or SARIMA.

The SARIMA Notation

- SARIMA is an extension of ARIMA that explicitly handles seasonality.
- The notation is a combination of two ARIMA models:
  - ARIMA(p,d,q)×(P,D,Q,m)
- Components:
- (p,d,q): The non-seasonal part of the model.
  - p: Non-seasonal Autoregressive order
  - d: Non-seasonal Differencing order
  - q: Non-seasonal Moving Average order
- (P,D,Q,m) : The seasonal part of the model.
  - P: Seasonal Autoregressive order
  - D: Seasonal Differencing order

Non-Seasonal vs. Seasonal Components

- The non-seasonal components (p,d,q) model the short-term dependencies in the data. They capture the relationships between adjacent observations.
- The seasonal components (P,D,Q) model the dependencies between observations separated by the seasonal period m. For example, they capture the relationship between January sales this year and January sales last year.
- Stationarity: Both parts of the model require the data to be stationary.
  - Non-seasonal differencing (d) removes the trend.
  - Seasonal differencing (D) removes the seasonal trend and mean.
  - Formal verification is often performed using the Augmented Dickey-Fuller (ADF) test.

Identifying Seasonality: Beyond Visuals

- While visual inspection of a time series plot is the first step, a more formal method is required.
- Key Tool: The Autocorrelation Function (ACF) plot.
- How it Works: An ACF plot shows the correlation between a time series and its lagged versions.
- Identifying Seasonality: For a seasonal series, the ACF plot will show large, significant spikes at multiples of the seasonal lag s.
- Example: For monthly data (m=12), you will see strong peaks at lags 12, 24, 36, and so on.

ACF Plot Example for Seasonal Data

- Example: Monthly Air Passengers dataset.
  - The plot shows strong autocorrelation with a slow decay at all lags, indicating a non-stationary series.
  - Crucially, it also shows very large spikes at lags 12, 24, and 36. This is a clear indication of a seasonal pattern with a period of 12.
  - This plot suggests the need for both non-seasonal and seasonal differencing

Seasonal Differencing (D)

- Seasonal differencing involves taking the difference between an observation and the corresponding observation from the previous season.
- The operation is expressed as: $\nabla(m)yt = yt - yt\text{-}m$
- This is different from regular differencing, which focuses on short-term trend removal: $\nabla yt = yt - yt\text{-}1$
- Purpose: Seasonal differencing removes the seasonal trend and mean, effectively making the series stationary with respect to the seasonal component.

Seasonal AR (P) and MA (Q) Terms

- Once a series is made stationary (both non-seasonal and seasonal), we can model the remaining autocorrelation with the seasonal AR and MA terms.
- Seasonal Autoregressive (P) term:
  - Models the correlation between an observation and a previous observation at a seasonal lag.
  - For example, with m=12, a P=1 term would model the correlation between sales in January 2025 and sales in January 2024.
- Seasonal Moving Average (Q) term:
  - Models the correlation between the current error and a previous error at a seasonal lag.
  - For a Q=1 term with m=12, it would model the relationship between the current forecast error and the error from 12 periods ago

A Step-by-Step Approach to SARIMA Modeling

- Modeling a SARIMA process requires a systematic approach.
- Step 1: Data Visualization: Plot the time series to identify seasonality, trend, and overall characteristics.
- Step 2: Differencing: Apply seasonal and/or regular differencing until the series is stationary. Use formal tests like the ADF test to confirm.
- Step 3: Parameter Identification: Use the ACF and PACF plots of the stationary series to identify the non-seasonal (p,q) and seasonal (P,Q) parameters.
- Step 4: Model Fitting: Fit the proposed SARIMA model using statistical software.
- Step 5: Diagnostics: Check the residuals for a good fit. The residuals should be uncorrelated and normally distributed.
- Step 6: Forecasting: Use the validated model to generate future forecasts.

Step-by-Step Parameter Identification

- Step 1: Determine m:
  - How: Visually inspect the original time series plot. Identify the length of the repeating cycle.
  - Example: Monthly data has m=12. Quarterly data has m=4.
- Step 2: Determine D:
  - How: Examine the ACF plot of the original data. If there are slow-decaying spikes at the seasonal lags (e.g., 12, 24, 36), a seasonal difference is needed. This indicates a seasonal trend.
  - Value: If you perform one seasonal difference, then D=1.
- Step 3: Determine d:
  - How: After applying seasonal differencing, examine the new ACF plot. If there are still slow-decaying spikes at non-seasonal lags (e.g., 1, 2, 3), a regular difference is needed.
  - Value: If you perform one regular difference, then d=1.

Identifying `p, q, P, Q` from Stationary Plots

- Now that the series is stationary, we analyze the ACF and PACF plots to identify the remaining terms.
- Non-seasonal terms (p,q):
  - Look for significant spikes at the non-seasonal lags (lags 1, 2, 3, etc.).
  - For q: Spikes in the ACF cut off after a few lags. The order q corresponds to the number of spikes.
  - For p: Spikes in the PACF cut off after a few lags. The order p corresponds to the number of spikes.
- Seasonal terms (P,Q):
  - Look for significant spikes at the seasonal lags (lags m, 2m, 3m, etc.).
  - For Q: A spike in the ACF at lag s that is not followed by other significant spikes at seasonal multiples. Q=1.
  - For P: A spike in the PACF at lag s that is not followed by other significant spikes at seasonal multiples. P=1.
- Tip: The seasonal terms often dominate the plots, so it's best to identify them first.

Model Fitting and Diagnostics

- Once a model is proposed, we fit it to the data using statistical software.
- Python Example: We will use the SARIMAX function from the statsmodels library.
- Diagnostics: A crucial final step to ensure the model is a good fit.
  - Residual Plot: The residuals (the difference between the actual values and the model's predictions) should look like white noise.
  - They should have a constant variance and a zero mean.
  - They should be randomly distributed with no discernible patterns.
  - Ljung-Box Test: A formal statistical test to check for autocorrelation in the residuals. A high p-value (e.g., > 0.05) indicates that the residuals are not significantly different from white noise, which is a good sign.

Maximum likelihood estimation

- Parameter Estimation
  - Once the model orders (p,d,q) (P,D,Q) are identified, we estimate the parameters
- Maximum Likelihood Estimation (MLE)
  - The statsmodel package uses MLE to find the parameter values that maximize the probability of observing the given data.
- Analogy to Least Squares
  - For ARIMA/SARIMA models, MLE is a technique similar to minimizing the sum of squared errors
- Software Differences
  - ARIMA/SARIMA models are complex to estimate, so different software packages may produce slightly different results due to variations in their estimation and optimization algorithms.

Information Criteria

- What are Information Criteria?
  - Metrics used to compare the quality of statistical models.
  - They provide a way to balance model fit (how well it explains the data) with model complexity (the number of parameters).
- Why use them for ARIMA models?
  - Help in objectively selecting the best model from a set of candidate models with different parameter values.
- The Goal:
  - A good model is one that minimizes the value of the information criterion.
  - The most common criteria are AIC, AICc, BIC, and HQIC.

Akaike's Information Criterion (AIC)

- Definition: A metric that estimates the relative amount of information lost by a given model.
- Formula:

  $AIC = -2\log(L) + 2(p+q+k+1)$
- Formula Breakdown:
  - L is the likelihood of the data from the estimated model.
  - p is the non-seasonal autoregressive order.
  - q is the non-seasonal moving average order.
  - k is a constant term: k=1 if a constant is included, and k=0 if it is not.
  - The last term, 2(p+q+k+1), represents a penalty for complexity (more parameters lead to a higher AIC).

Corrected AIC (AICc), Bayesian Information Criterion (BIC)

- Corrected AIC (AICc)
  - An improved version of AIC, particularly for small sample sizes.
  - It adds a heavier

$$AIC_c = AIC + \frac{2(p+q+k+1)(p+q+k+2)}{T-p-q-k-2}$$

of data is limited.
  - Formula:

- Bayesian Information Criterion (BIC)
  - Similar to AIC bu

$$BIC = AIC + [\log(T) - 2](p+q+k+1)$$

  - It places a much stronger penalty on model complexity.
  - Formula:

Hannan-Quinn Information Criterion (HQIC)

- A criterion that provides a penalty between AIC and BIC.
- It is often considered consistent, meaning it will select the true model order if one exists, unlike AIC.
- Formula:
  - $HQIC = -2\log(L) + 2(p+q+k+1)\log(\log(T))$
- Choosing a Criterion
- Prefer to use the AICc as it is a more reliable guide for model selection.

A Critical Limitation

- Warning
  - Information criteria are not good for selecting the order of differencing (d).
- Why?
  - Differencing fundamentally changes the data on which the likelihood is calculated. This makes the AIC or BIC values of models with different differencing orders incomparable.
- Correct Approach
  - First, determine the appropriate order of differencing (d) through visual inspection of plots (ACF) and formal tests (ADF).
  - Once d is set, use the AICc to select the optimal values for the autoregressive (p) and moving average (q) terms.